Engineering & Technology — Embedded Systems

Edge-AI for Early Detection of *Sugarcane Woolly Aphid* Infestation

Embedded & IoT Systems

On-device inference · Dual-sensor fusion
Solar-powered · Fully offline

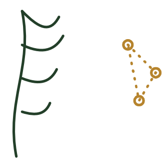
A standalone embedded system that listens for wingbeats and watches leaf colour in the field — flagging infestations before they spread, without a single byte sent to the cloud.

The Problem Introduction & Relevance

Sugarcane underwrites the rural economy across the Pune–Ahmednagar belt, and woolly aphid colonies can spread through a plot within days of the first humid spell. By the time white cottony patches are visible from the field bund, the infestation is already past easy control.

Scouting still happens on foot, plot by plot, which means detection lags the pest by days rather than hours.

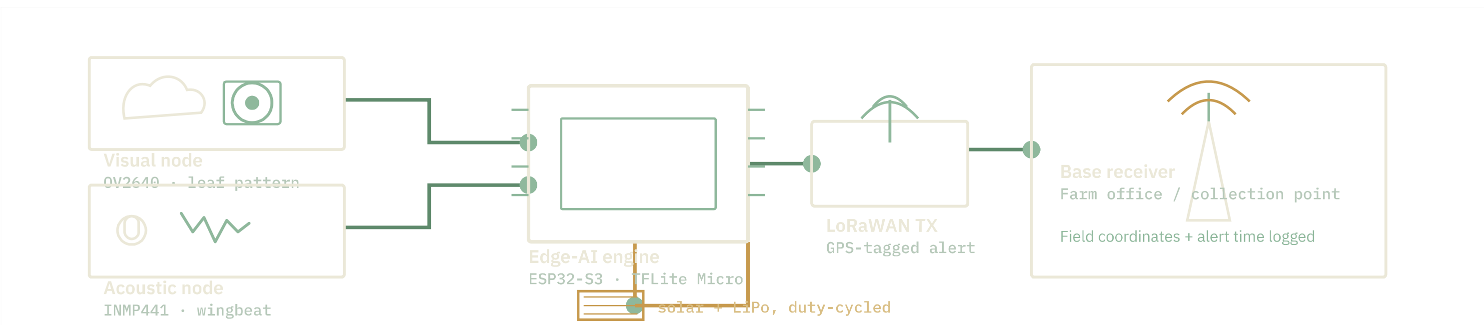
Engineering gap



Woolly aphid pressure builds fastest in **dense, interior plantings** with limited road or signal access — exactly where field staff spend the least time and connectivity is weakest.

System Architecture Methodology

Two sensor nodes feed a single low-power controller. Inference runs entirely on-chip; only a short alert packet ever leaves the device.



Building the Model 20

Paired field dataset: healthy vs. infested leaf imagery, shot across affected plots at varied light conditions.

Wingbeat audio isolated from field wind and machinery noise, labelled against confirmed colony sightings.

Deep-sleep duty cycling between capture windows, sized against a small solar-trickle battery budget.

`ESP32-S3`	Processing hub
`OV2640`	Leaf imaging
`INMP441`	Wingbeat capture

Design Effort 20

TensorFlow Lite Micro model quantized and flashed directly to the controller — no round-trip to a server.

Sensor housing and mount iterated across three field placements to reduce wind noise and glare on the lens.

Power draw profiled per duty cycle to size the solar panel and battery for unattended field operation.

`TFLite Micro`	On-chip inference
`LoRaWAN`	Alert transmit
`Solar + LiPo`	Field power

What's New 20

i.

Dual-fusion detection

Visual and acoustic signals are cross-checked before an alert fires — a moving leaf or a passing insect alone won't trigger a false positive.

ii.

Fully offline computing

Zero cellular data, zero subscription. The device is as useful at the edge of a signal map as it is beside a tower.

Field Performance Outcomes — 20

94.6%

On-chip inference accuracy, achieved directly on the controller during lab trials — no post-processing on a separate machine.

~₹1,800

Target build cost per unit — sized for a smallholder farm budget, not an estate-scale procurement.

Duty-cycled power draw

Sense

Transmit

Sleep

Relative draw across the sense–sleep–transmit cycle, used to size the solar-trickle battery.

Why it holds up in the field

No connectivity dependency, no recurring cost, and a detection window measured in hours instead of days — built around how cane is actually farmed in interior Maharashtra plots, not around the infrastructure that's missing there.

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